

Leveraging Machine Learning and Natural Language Processing for Detecting Coordinated Disinformation Campaigns on Social Media Platforms

Khaled Ahmed

2026

Abstract

Social media has made it possible for coordinated disinformation campaigns to reach and manipulate public opinion at a scale that traditional propaganda never could. This paper reviews research on machine learning (ML) and natural language processing (NLP) approaches to detecting such campaigns, with a focus on the X (Twitter) platform. Drawing on research articles, benchmark datasets, and case studies published between 2016 and 2026, it examines classical and deep learning methods for identifying coordinated activity, compares their relative effectiveness, and considers what network-level approaches add beyond content analysis alone. The review pays close attention to the features that distinguish coordinated behavior from organic activity, the trade-offs between different algorithmic approaches, and the ethical concerns that arise when such systems are deployed at scale. Across the literature examined, hybrid approaches that combine content and network features consistently outperform single-method systems, though significant limitations around generalization, interpretability, and fairness remain.

Keywords: disinformation detection, machine learning, natural language processing, social network analysis, bot detection, coordinated inauthentic behavior

1. Introduction

1.1 Background

Social media networks have transformed the speed and scale of information dissemination over the past two decades, and not always for the better. One of the clearest downsides is the volume of disinformation these platforms now carry. What distinguishes coordinated disinformation campaigns from the ordinary, unsystematic spread of false claims by individual users is organization: campaigns of this kind rely on bots, cyborg accounts, and paid human operators working together to reach far larger audiences than any single account could manage on its own. The accounts involved may act independently, but past research has also documented cases in which they were directed by a single coordinating entity.

1.2 The Scale of the Problem

Traditional media placed a natural ceiling on propaganda: there were only so many broadcasters, newspapers, and other carriers through which false information could travel. Social media removed that ceiling. A small group of individuals can now amplify a message many times over using networks of bots, and the resulting volume of disinformation has grown accordingly. These campaigns are not confined to any one type of platform; they have appeared on both privately owned and state-linked networks. The scale involved is considerable: Twitter alone disclosed information operation accounts spanning seventeen countries, involving more than fifty thousand accounts identified between 2007 and 2020 (Smith et al., 2021), while researchers working independently have identified hundreds of thousands of additional bot accounts within dedicated bot-detection benchmark datasets (e.g., Feng et al., 2021).

1.3 Why Machine Learning and NLP?

Manual review simply cannot keep pace with the volume involved, which is why researchers have shifted toward models that can recognize bot activity automatically. The solutions proposed so far span a range of technical approaches. Earlier work leaned on traditional machine learning algorithms such as Support Vector Machines, Random Forests, and Long Short-Term Memory recurrent neural networks; more recent studies have turned to transformer-based language models such as BERT. A separate strand of research takes a different angle altogether, using network-based algorithms that examine how accounts interact with one another rather than analyzing post content in isolation.

1.4 Problem Statement

Despite this progress, detecting coordinated disinformation campaigns remains genuinely difficult. Part of the problem is adversarial: actors can simply change their behavior once they suspect they are being monitored, which puts detection systems in a constant game of catch-up. Traditional machine learning models add a further constraint, since they typically need large amounts of training data to perform well. False positives compound the difficulty further, since legitimate accounts, including those belonging to journalists, are occasionally caught up in detection efforts. Taken together, these issues point to a need for a clearer understanding of which technologies actually work, and just as importantly, where and why current approaches still fall short.

1.5 Purpose of the Study

This study examines how machine learning and natural language processing (NLP) approaches have been applied to identify coordinated disinformation campaigns, with particular attention to campaigns on the Twitter/X platform between 2016 and 2026. Rather than proposing a new detection method of its own, it synthesizes existing research in order to build a clearer picture of which approaches are most promising, and under what conditions.

2. Research Questions

This study evaluates the efficacy of machine learning and natural language processing techniques in this context. It is guided by the following main question and five sub-questions:

Main Research Question: To what extent can machine learning and natural language processing techniques accurately detect and classify coordinated disinformation campaigns, as opposed to organic user-generated content?

RQ1: What features most distinguish coordinated bot or troll activity from genuine human interaction?

RQ2: Which machine learning models achieve the highest accuracy, precision, and recall in classifying disinformation content?

RQ3: How effective is Social Network Analysis in detecting coordinated inauthentic behavior compared to purely content-based approaches?

RQ4: What are the key limitations and ethical risks associated with automated disinformation-detection systems?

RQ5: How do currently deployed detection systems perform against documented real disinformation campaigns?

3. Methodology

This paper is structured as an evidence-based literature review rather than an original experimental study. It synthesizes findings from peer-reviewed articles, preprints, benchmark dataset papers, and documented case studies concerned with machine learning and natural language processing approaches to detecting coordinated disinformation on social media, with particular attention to the X (Twitter) platform. The sources examined were selected for their direct relevance to the five research questions guiding this study: the features that distinguish coordinated behavior, the comparative performance of different algorithmic approaches, the contribution of Social Network Analysis, the limitations and ethical risks associated with automated detection, and the performance of detection systems against documented real-world campaigns. Consistent with the scope outlined in the Introduction, the review draws on research published between 2016 and 2026.

The literature reviewed spans classical machine learning methods, such as Support Vector Machines, Random Forests, and Naive Bayes; deep learning and transformer-based approaches, including Long Short-Term Memory networks and BERT-derived architectures (Kula et al., 2022); and network-based detection frameworks such as VigDet (Zhang et al., 2021). It also draws on established benchmark datasets, including TwiBot-20 (Feng et al., 2021) and FakeNewsNet (Shu et al., 2020), and on multilingual evaluation work such as the PolyTruth corpus (Gouliev et al., 2025), all of which are discussed in detail in the sections that follow. Rather than applying a formal systematic-review protocol, this study organizes and compares the identified sources thematically around each research question, examining where studies converge, where their approaches or findings diverge, and what these patterns indicate about the current state of coordinated disinformation detection. This comparative synthesis, rather than any new data collection or experimentation, forms the analytical basis for the discussion and conclusions presented later in the paper.

4. Theoretical Framework

4.1 Big Data in Social Media Analysis

Social media platforms generate data defined by the three Vs of Big Data: Volume, Velocity, and Variety. Detecting coordinated disinformation means processing this data at a scale manual review simply cannot reach, which is why most detection systems in the field are built on Big Data infrastructure such as distributed storage and streaming APIs.

4.2 Machine Learning Approaches

Detection systems generally fall into two broad categories. Classical supervised learning approaches, such as Support Vector Machines, Random Forests, and Naive Bayes, rely on hand-engineered features such as posting frequency, account age, and follower-to-following ratios. They are computationally efficient, but limited in how much contextual nuance they can capture. Deep learning approaches, including Long Short-Term Memory networks and transformer-based models such as BERT, tend to achieve higher accuracy, but at a cost: it is often difficult to understand why a given model flagged a particular account, a limitation that matters more once these systems move from research settings into real deployment.

4.3 Natural Language Processing

NLP techniques allow systems to analyze what an account posts, not just how it behaves. Two applications stand out in the literature: semantic similarity detection, which can surface the near-duplicate messages that are a hallmark of coordinated campaigns, and sentiment or stance detection, which can expose artificially amplified emotional framing.

4.4 Social Network Analysis

Coordination is not always visible in what accounts post. It can also show up as unusually dense clusters of interaction, synchronized posting times, or shared use of the same third-party scheduling tools, patterns that content analysis alone would miss. Graph-based approaches address this by modeling accounts as nodes and interactions as edges, which makes it possible to detect clusters that behave like an orchestrated network rather than an organic audience.

4.5 The Bot-Human Spectrum

Not all inauthentic activity is automated. Research distinguishes between fully automated bots, cyborg accounts that combine automation with occasional human intervention, and paid human operators such as troll farm employees. This distinction matters for detection: methods built to catch purely behavioral bot signatures may miss human-operated coordinated accounts entirely, which is one of the reasons hybrid approaches that combine content and network detection have gained traction.

5. Literature Review and Applied Analysis

5.1 RQ1: Feature-Based Detection of Coordinated Behavior

Smith et al. (2021) offer a broad approach to RQ1 by combining natural language processing, machine learning, and graph analytics to examine disinformation narratives and influential actors. Their work illustrates the value of integrating multiple forms of evidence rather than relying on activity counts alone. Alizadeh et al. (2020) approach the problem from a different angle, showing that content-based features derived from what accounts post can distinguish coordinated influence-operation content from organic activity across multiple documented troll campaigns. Taken together, these studies suggest that coordinated behavior can leave detectable signals in both network structure and content, although the usefulness of particular features may vary across campaign types and operational contexts.

Keller et al. (2020) approach the same question from a narrower but complementary angle, focusing specifically on political astroturfing rather than disinformation broadly. Their finding that coordinated campaigns tend to rely on a small number of human agents producing large volumes of highly similar content within short time windows points to a feature set distinct from the network-community signals Smith et al. (2021) rely on: topic similarity, word count distribution, and shared linked articles. Read together, these studies suggest that the most distinguishing features of coordinated behavior are not uniform across campaign types: network-community structure may be more diagnostic for large, state-linked operations, content-based signals can generalize across troll campaigns regardless of network position, and content-similarity metrics may better capture smaller, human-driven astroturfing efforts.

5.2 RQ2: Algorithmic Performance across Model Types

A comparison across model types shows a fairly consistent performance gradient. Classical methods such as Naive Bayes and Support Vector Machines produce respectable but not exceptional results on tasks such as COVID-19 disinformation detection: Olayiwola et al. (2023), for example, report a Support Vector Machine classifier reaching a 96 percent F1-score on a COVID-19 fake-news dataset, a stronger showing than the decision-tree, random-forest, and logistic-regression baselines evaluated in the same study. Deep learning methods, and transformer-based models in particular, tend to push accuracy noticeably higher: Kula et al. (2022) find that BERT-derived architectures can outperform earlier convolutional and recurrent neural-network baselines on disinformation classification tasks. BERT-derived architectures have also demonstrated strong performance in fake-news classification tasks, although reported results vary substantially depending on the dataset, language, and evaluation setting. That gradient does not hold, however, once language enters the picture. Multilingual evaluation using the PolyTruth corpus found that RemBERT maintained strong performance even in low-resource languages, while mBERT and XLM degraded considerably when training data was scarce (Gouliev et al., 2025). The implication is that reported accuracy figures say less about a model's intrinsic capability than about the language and dataset it was evaluated on, a caveat that complicates any straightforward ranking of 'best' models across the literature.

5.3 RQ3: Network-Based Detection Approaches

Network structure, on its own, carries detection signals that content analysis cannot reach. VigDet, developed at the University of Southern California, illustrates this well: Zhang et al. (2021) apply a knowledge-informed neural temporal point process to model the timing and structure of coordinated posting behavior, directly addressing the sparsity problem that limits purely activity-based detectors. Emeric and Victor (2024) extend the same logic further, proposing multi-layer network models that link accounts across several platforms at once to surface coordinated clusters that would stay invisible if any single platform were analyzed in isolation. Ng and Iamnitshi (2023) make a related point from a different angle, arguing that combining network, temporal, and content dimensions into a single framework can provide a broader representation of the diverse nature of coordinated activity than relying on any one signal alone. What these approaches share is a willingness to treat structure, rather than content, as central evidence of coordination, and the fact that this works reinforces a point raised earlier in the theoretical framework: network-based and content-based detection methods are complementary rather than competing options.

5.4 RQ4: Limitations, Bias, and Ethical Considerations

High accuracy figures can obscure real limitations. Menzner and Leidner (2024), evaluating pre-trained neural transformers on the task of news bias detection, find that no automated system fully eliminates false positives and false negatives, particularly in cases where judging a passage correctly requires context beyond the text itself. A case study reviewed under the OECD Catalogue of Tools and Metrics for Trustworthy AI examined a BERT-based disinformation classifier trained on the Twitter1516 dataset and found a measurable, though contextually justifiable, difference in treatment across user subgroups (OECD.AI Policy Observatory, 2024), a reminder that algorithmic bias remains a live concern even in models that otherwise perform well. Interpretability compounds the problem: deep learning models are typically opaque enough that platform operators, regulators, and affected users alike struggle to understand why a given account or post was flagged. That opacity is not a minor technical inconvenience when these systems inform decisions such as account suspension or content removal; it raises important questions about accountability and due process when errors cannot be readily explained or reviewed. Applied systems built for real-world deployment illustrate the practical stakes of these limitations: Cartwright et al. (2022) developed a machine-learning and deep-neural-network pipeline, commissioned by the Canadian Department of National Defence, to identify hostile disinformation activity in near-real-time, underscoring that the same interpretability and bias concerns arise the moment such systems move from benchmark datasets into operational settings.

5.5 RQ5: Evaluation Against Documented Real-World Campaigns

Benchmark datasets provide the empirical ground truth against which detection systems are evaluated, and TwiBot-20 illustrates why this matters. Feng et al. (2021) built the dataset from over 229,000 users and more than 33 million tweets specifically to correct the low user diversity and limited information of earlier benchmarks, and in doing so exposed a notable weakness: when they tested previously competitive bot-detection methods against it, several failed to replicate their claimed performance. This finding is significant, as it suggests some of the field's earlier reported successes may have said as much about the limitations of the benchmarks used as about the methods themselves. Saeed et al. (2024) reinforce this generalization concern from another angle: a classifier they trained on traits shared across nineteen state-sponsored disinformation campaigns on Twitter correctly identified up to 94 percent of accounts from campaigns it had not previously seen, suggesting that campaign-agnostic feature design may partly mitigate the generalization problem that TwiBot-20 exposes. FakeNewsNet, built from fact-checked content on PolitiFact and GossipCop (Shu et al., 2020), offers a related but distinct perspective, having been used to study how verified false content spreads through user-sharing behavior. More recent work continues to build on this foundation: Cui et al. (2025) use FakeNewsNet, alongside the LIAR and COVID-19 datasets, to evaluate a multi-tool large language model agent for verifiable misinformation detection, illustrating that these benchmark resources remain central even as detection techniques move beyond classical supervised classifiers. Taken together, these resources point toward the same conclusion reached elsewhere in this review: detection systems tend to perform well on the datasets they were tuned for, but generalizing to genuinely new, previously unseen campaigns remains a largely unresolved problem.

6. Discussion and Key Challenges

Reading across this literature rather than through it section by section, a consistent pattern emerges: no single technique, whether purely content-based, purely behavioral, or purely network-based, solves disinformation detection on its own. Several of the stronger systems discussed in the studies reviewed here combine multiple signal types, using NLP to assess content similarity, behavioral features to flag suspicious account activity, and network analysis to expose coordination that would be invisible at the individual-account level. This is not a coincidence; it follows directly from the bot-human spectrum discussed in the theoretical framework. Purely behavioral detection may miss human-operated coordinated accounts, while purely content-based detection may miss coordination that manifests structurally rather than semantically; combining these signals can therefore provide a broader basis for detection.

Three challenges recur across the studies examined and, taken together, define much of what remains unresolved in the field. Generalization is the most immediate: models trained on historical campaigns, including those associated with well-documented influence operations, tend to underperform against new and previously unseen coordinated activity, a pattern the TwiBot-20 evaluation results (Feng et al., 2021) make concrete rather than theoretical. Interpretability is the second, and it sits in tension with the first, since the deep learning models that generalize best are typically also the hardest to explain, which is a significant limitation in contexts such as platform moderation or regulatory oversight, where accountability is essential. Fairness is the third, and arguably the most consequential in practice: even systems with strong aggregate accuracy have been shown to treat user subgroups differently, which raises ethical questions that accuracy metrics alone cannot answer. None of the studies reviewed here offers a way to resolve all three challenges at once, and the literature as a whole gives little indication that doing so will be straightforward.

7. Conclusion and Recommendations

This study set out to examine how well machine learning and natural language processing techniques can detect coordinated disinformation campaigns on social media. The evidence gathered here supports a fairly clear answer: individual techniques, whether content-based, behavioral, or network-based, do achieve meaningful results on their own, but hybrid systems that combine multiple signal types often demonstrate advantages, particularly when addressing complex forms of coordinated behavior and when combining complementary evidence from content, behavior, and network structure.

These findings point to four practical recommendations:

First, future detection systems should be built around hybrid architectures that combine NLP-based content analysis with graph-based network analysis, rather than relying on either approach in isolation.

Second, researchers and platforms should invest in explainable AI techniques alongside deep learning-based classifiers, so that flagged accounts or content can be reviewed and justified transparently rather than accepted without explanation.

Third, benchmark datasets should be continually updated and diversified across languages and platforms, given the documented variation in model performance tied to linguistic resource

availability.

Finally, any large-scale deployment of automated disinformation detection should incorporate systematic bias auditing, since differential treatment across user subgroups has been documented even in otherwise high-accuracy models.

Taken as a whole, the literature reviewed here suggests that the field has moved past asking which single technique works best, and toward the more useful question of how different techniques can be combined responsibly. That shift, more than any individual accuracy figure, may be the most durable finding of this review.

References

- Alizadeh, M., Shapiro, J. N., Buntain, C., & Tucker, J. A. (2020). Content-based features predict social media influence operations. *Science Advances*, 6(30), eabb5824. <https://doi.org/10.1126/sciadv.abb5824>
- Cartwright, B., Frank, R., Weir, G., & Padda, K. (2022). Detecting and responding to hostile disinformation activities on social media using machine learning and deep neural networks. *Neural Computing and Applications*, 34(18), 15141–15163. <https://doi.org/10.1007/s00521-022-07296-0>
- Cui, Z., Huang, T., Chiang, C.-E., & Du, C. (2025). Toward verifiable misinformation detection: A multi-tool LLM agent framework. *arXiv*. <https://arxiv.org/abs/2508.03092>
- Emeric, A., & Victor, C. (2024). Interpretable cross-platform coordination detection on social networks. In *Advances in Complex Networks and Their Applications* (pp. 143–155). Springer. https://doi.org/10.1007/978-3-031-53503-1_12
- Feng, S., Wan, H., Wang, N., Li, J., & Luo, M. (2021). TwiBot-20: A comprehensive Twitter bot detection benchmark. In *Proceedings of the 30th ACM International Conference on Information & Knowledge Management* (pp. 4485–4494). Association for Computing Machinery. <https://doi.org/10.1145/3459637.3482019>
- Gouliev, Z., Waters, J., & Wang, C. (2025). PolyTruth: Multilingual disinformation detection using transformer-based language models. *arXiv*. <https://arxiv.org/abs/2509.10737>
- Keller, F. B., Schoch, D., Stier, S., & Yang, J. (2020). Political astroturfing on Twitter: How to coordinate a disinformation campaign. *Political Communication*, 37(2), 256–280. <https://doi.org/10.1080/10584609.2019.1661888>
- Kula, S., Kozik, R., & Chora, M. (2022). Implementation of the BERT-derived architectures to tackle disinformation challenges. *Neural Computing and Applications*, 34(23), 20449–20461. <https://doi.org/10.1007/s00521-021-06276-0>
- Menzner, T., & Leidner, J. L. (2024). Experiments in news bias detection with pre-trained neural transformers. In *Proceedings of the 29th European Conference on Information Retrieval (ECIR 2024)* (Vol. 4, pp. 270–284). Springer. https://doi.org/10.1007/978-3-031-56066-8_22
- Ng, K., & Iamnitchi, A. (2023). Coordinated information campaigns on social media: A multifaceted approach. In *Disinformation in Open Online Media (MISDOOM 2023)* (Lecture Notes in Computer Science, Vol. 14397, pp. 103–118). Springer. https://doi.org/10.1007/978-3-031-47896-3_8
- OECD.AI Policy Observatory. (2024). Higher-dimensional bias in a BERT-based disinformation classifier. Catalogue of Tools & Metrics for Trustworthy AI. <https://oecd.ai/en/catalogue/tools/unsupervised-bias-detection-tool/tool-use-cases/higher-dimensional-bias-in-a-bert-based-disinformation-classifier>
- Olayiwola, A., Oyediji, A. O., Omoyeni, O., Ayemimowa, O., & Olaoluwa, M. (2023). Comparative analysis of machine learning models for detection of fake news: A case study of Covid-19. *JITCE (Journal of Information Technology and Computer Engineering)*, 7(1), 29–33. <https://doi.org/10.25077/jitce.7.01.29-33.2023>
- Saeed, M. H., Ali, S., Paudel, P., Blackburn, J., & Stringhini, G. (2024). Unraveling the web of disinformation: Exploring the larger context of state-sponsored influence campaigns on Twitter. In *Proceedings of the 27th International Symposium on Research in Attacks, Intrusions and Defenses (RAID 2024)* (pp. 353–367). Association for Computing Machinery. <https://doi.org/10.1145/3678890.3678911>
- Shu, K., Mahudeswaran, D., Wang, S., Lee, D., & Liu, H. (2020). FakeNewsNet: A data repository with news content, social context, and spatiotemporal information for studying fake news on social media. *Big Data*, 8(3), 171–188. <https://doi.org/10.1089/big.2020.0062>

- Smith, S. T., Kao, E. K., Mackin, E. D., Shah, D. C., Simek, O., & Rubin, D. B. (2021). Automatic detection of influential actors in disinformation networks. *Proceedings of the National Academy of Sciences*, 118(4), e2011216118. <https://doi.org/10.1073/pnas.2011216118>
- Zhang, Y., Sharma, K., & Liu, Y. (2021). VigDet: Knowledge informed neural temporal point process for coordination detection on social media. *Advances in Neural Information Processing Systems*, 34, 3218–3231.